

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

<reasoning process here> The boy presents with a constellation of symptoms including poor concentration , hyper activity , impuls ivity , forget fulness , and disruptive behavior in the classroom . He has also had a history of aggression when frustrated . Development al milestones are met , and he is otherwise healthy . The history of recurrent ear infections is not clearly relevant given the lack of hearing loss or indication of sensory impairment . There is no physical examination finding to explain the symptoms .

The father has Graves ' disease , which is a family history of autoimmune thyroid disorder . Although the boy has not been evaluated for thyroid function , the key consideration is that hyperthyroidism can manifest with irritability , hyper activity , poor concentration , and emotional lability âĖŁ symptoms that mimic attention deficit hyper activity disorder (ADHD), especially in children .

While behavioral therapy (A) is a common first-line treatment for ADHD , particularly in cases with clear behavioral dysregulation , the presence of a maternal history of major depression and a paternal history of Graves âĖŁ disease raises concern for a possible endocrine cause . Moreover , the history of aggression and impuls ivity âĖŁ and an otherwise healthy child who has reached all developmental milestones âĖŁ increases suspicion for a possible endocrine cause like hyperthyroidism .

In children with suspected hyperthyroidism , especially those with hyper activity , irritability , and difficulty concentrating , it is essential to rule out hyperthyroidism , particularly in the context of a family history of Graves âĖŁ disease . Methimazole (B), a medication used to treat hyperthyroidism , should be considered if hyperthyroidism is confirmed . Fluoxetine (C) is an antidepressant used in treating anxiety and depression , which might be relevant in the maternal psychiatric history , but in absence of any primary depression symptoms , it is not the initial step . Hearing aids (D) would only be indicated for sensorineural hearing loss and are not indicated in a child with no hearing issues .

Therefore , the most appropriate next step is to assess and evaluate for possible hyperthyroidism , which should be done with thyroid function tests , such as TSH , free T4 .

If hyperthyroidism is found , treatment with methimazole (a thionamide) would be indicated to control thyroid hormone levels . However , the initial diagnostic step is testing .

Thus , the most appropriate next step is to perform thyroid function tests . But among the given answer choices , the only treatment options provided are A , B , C , D .

Note : The question asks for the most appropriate *next step in treatment* , not evaluation or testing .

Since therapy options are listed , if it is not possible to access testing , behavior therapy (A) is often considered for behavioral symptoms . But in a child with potential endocrine etiology , especially with suspected hyperthyroidism , the most appropriate treatment step initially is ruling out the underlying cause .

However , given the answer options , and since no testing option is available , and the question says " treatment " âĖŁ we interpret which treatment is most appropriate after diagnosis .

But the standard rule is : evaluate for possible diagnosis first , then treat .

Therefore , even though testing is more appropriate , if we must pick a treatment from options , and because the question says " treatment " next step , then the omission of thyroid testing is an error . However , among listed options , if we have to pick from treatment , then behavior therapy is first-line in symptom management . But the correct answer would only be after considering hyperthyroidism .

BUT âĖŁ in real pediatrics exam settings , especially in high-yield conditions , if a child has behavioral symptoms with family history of autoimmune endocrine disease , the treatment considered